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SP22-BSE-122

COMSATS University Islamabad (Lahore Campus)

■Midterm Exam ☐Terminal Examination – FALL 2023

Course Title:	Computer Networks	Course Code:	CSC340	Credit Hours:	4(3,1)
Course Instructor/s:	Dr. Shahbaz Akhtar Abid	Programme Name:	BS SE		
Time Allowed:	90 min	Maximum Marks:		50	

Important Instructions / Guidelines:

Question No 1. **Marks:4+10+6+5=25**

CLO: <1-2-3>; Bloom Taxonomy Level: <Understanding, Applying, Analyzing>

a) How does head-of-line (HOL) blocking problem affect the performance of HTTP1.1 for various web objects? Explain its impact using an example scenario and illustrate its solution incorporated in HTTP 2.0. (4)

b) Assume within your Web browser you click on a link to obtain a Web page. The IP address for the associated URL is not cached in your local host, so a DNS lookup is necessary to obtain the IP address. Suppose that n DNS servers are visited before your host receives the IP address from DNS; the successive visits incur an RTT of $RTT_1, \ldots, RTT_n$. Further suppose that the base web page associated with the link contains exactly one object, consisting of a small amount of HTML text. Let RTT_0 denote the RTT between the local host and the server containing the object. If the base HTML file references eight very small objects on the same server. Neglecting transmission times, how much time elapses with

i. Non-persistent HTTP with no parallel TCP connections? (3)

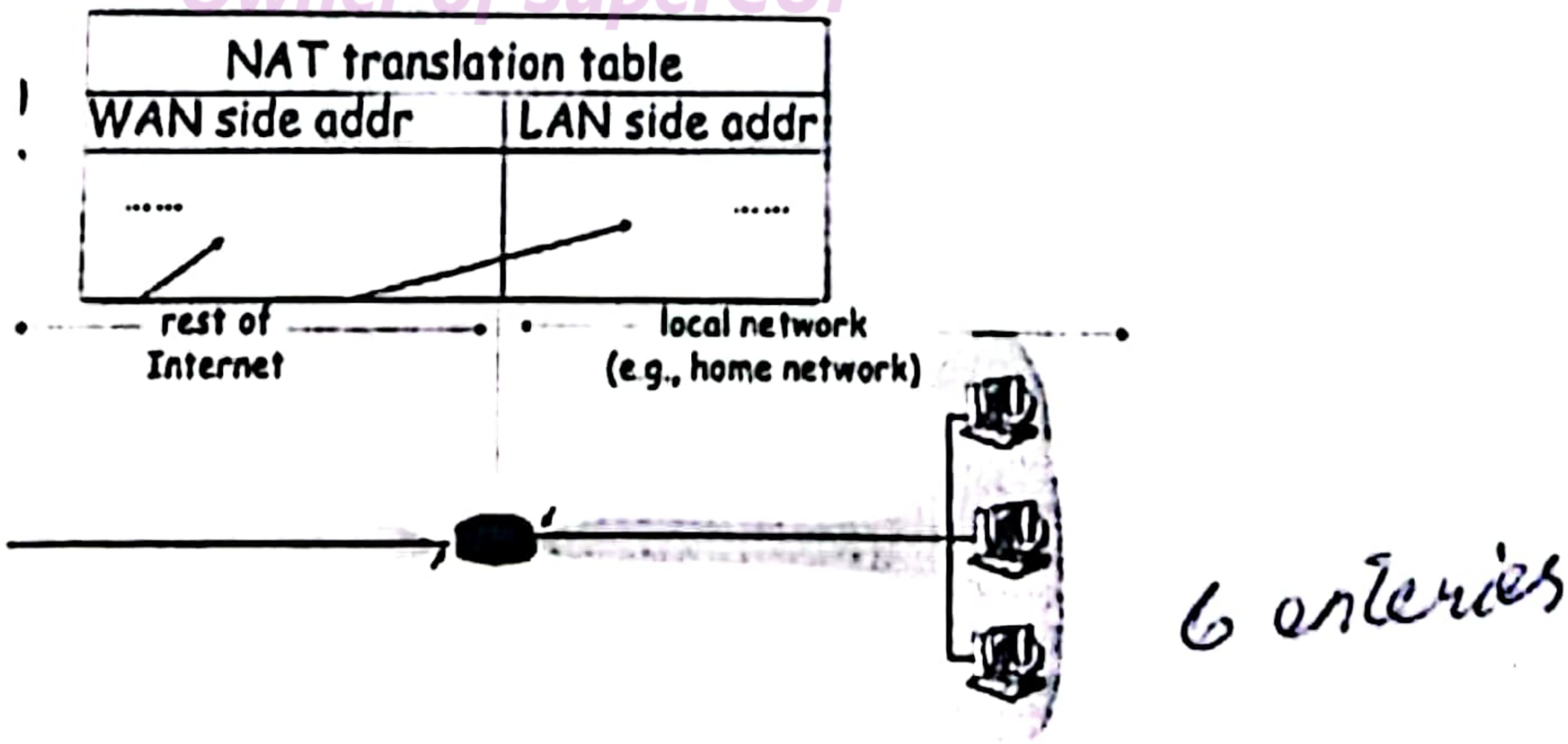
ii. Non-persistent with eight parallel connections? (4)

iii. Persistence HTTP with pipelining? (3)

c) How the denial-of-Service attack is generated using TCP SYN request messages and how to avoid? (6)

d) Consider a given network setup, assume the ISP assigns an IP address 138.76.29.7. The LAN network address is 10.0.0.0/24.

Compiled by Hafiz Ahmad
Owner of SuperCUI



Reproduce the above Figure by adding the following:

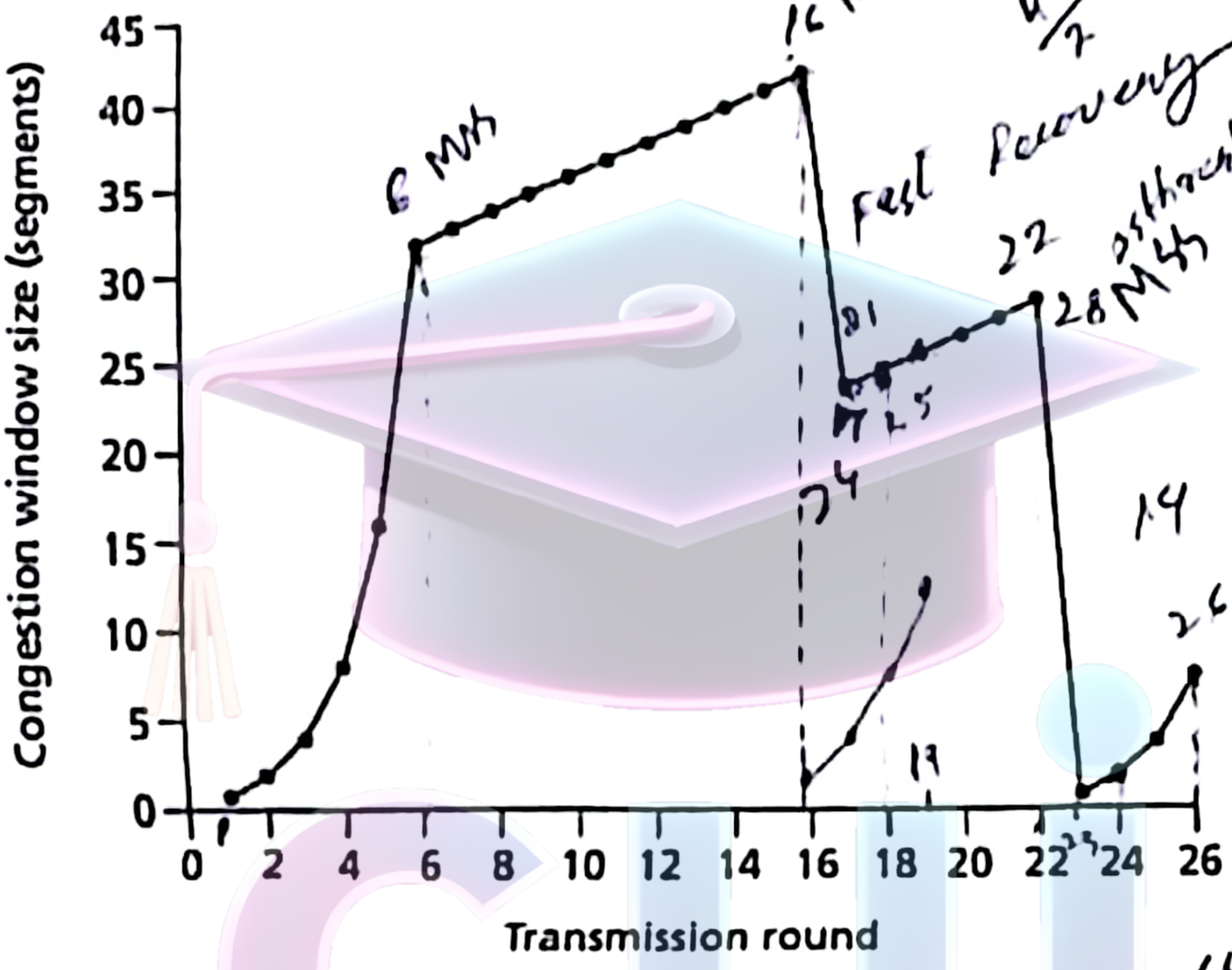
i. Assign IP addresses to all the interfaces in the LAN

ii. Assume each host in the LAN opens two TCP connections to 110.110.110.10 at port 80. Populate NAT table with mapping entries for the above scenario. (5)

Question No 2. **Marks: 2 x 8 = 16**

CLO: <1-2-4>; Bloom Taxonomy Level: <Remembering, Understanding, Analyzing>

Consider the following figure showing TCP window size as a function of time. Assume TCP Reno is the protocol experiencing the behavior shown in the figure, answer the following questions. In all cases, you should provide a short discussion justifying your answer.



1) Identify the intervals of time when TCP slow start is operating.
2) Identify the intervals of time when TCP congestion avoidance is operating.
3) After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
4) After the 22nd transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
5) What is the initial value of ssthresh at the first transmission round?
6) What is the value of ssthreshat the 18th transmission round?
7) What is the value of ssthreshat the 24th transmission round?
8) Suppose TCP Tahoe is used (instead of TCP Reno), and assume that triple duplicate ACKs are received at the 16th round. What are the ssthresh and the congestion window size at the 19th round?

Question No 3. **Marks: 9**

CLO: <1-2-3>; Bloom Taxonomy Level: <Remembering, Understanding, Applying>

Draw sender side finite state machine (FSM) of the reliable data transfer protocol (rdt) version 2.2

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